

2313666

殷家兴 自动化

电机①

2.1 解:

$$(1) T_F = F \cdot \frac{d}{2} = \cancel{300 \text{ N}\cdot\text{m}} 150 \text{ N}\cdot\text{m}$$

$$P = \frac{T_F \cdot \omega}{60} = \frac{T_F \cdot 2\pi n}{60} = 14500 \text{ W}$$

$$n_f = \frac{n}{j_1 j_2 j_3} = \frac{1450}{6} \text{ r/min}$$

$$P_f = T_F \cdot \frac{2\pi n_f}{60} = 3796.1 \text{ W}$$

$$(2) P_0 = \frac{P_f}{\eta^3} = \frac{P_f}{0.9^3} = 5207.3 \text{ W}$$

$$(3) GD^2 = GD_a^2 + \frac{GD_b^2}{j_1^2} + \frac{GD_c^2}{(j_1 j_2)^2} + \frac{GD_d^2}{(j_1 j_2 j_3)^2}$$

$$= 3.5 \text{ N}\cdot\text{m}^2 + 0.5 \text{ N}\cdot\text{m}^2 + 0.3 \text{ N}\cdot\text{m}^2 + 0.25 \text{ N}\cdot\text{m}^2$$

$$= 4.55 \text{ N}\cdot\text{m}^2$$

(4) 折算到电动机轴上的负载转矩

$$T_F = \frac{T_f}{j_1 j_2 j_3} = \frac{T_f}{j_1 j_2 j_3 j_3^2} = 34.3 \text{ N}\cdot\text{m}$$

$$\begin{cases} T_1 - T_{L1} = J \frac{dn}{dt} \\ T_{L1} = T_F \end{cases}$$

由于切削时电动机转速恒定, 因此 $\frac{dn}{dt} = 0$ 电动机的电磁转矩 $T_1 = T_F = 34.3 \text{ N}\cdot\text{m}$

$$(5) \text{机械损耗: } \Delta T = \frac{T_f}{j_1 j_2 j_3} - \frac{T_f}{j_1 j_2 j_3} = 9.3 \text{ N}\cdot\text{m}$$

电动机未切削时 $T_{L2} = 0 + \Delta T = 9.3 \text{ N}\cdot\text{m}$

$$\text{其电磁转矩为: } T_2 = T_{L2} + \frac{GD^2}{375} \frac{dn}{dt}$$

$$= 9.3 \text{ N}\cdot\text{m} + \cancel{9.7 \text{ N}\cdot\text{m}} 9.7 \text{ N}\cdot\text{m}$$

$$= 19.0 \text{ N}\cdot\text{m}$$